ELSEVIER
Journal of Public Economics 89 (2005) 351–380
JOURNAL OF PUBLIC ECONOMICS
www.elsevier.com/locate/econbase

School vouchers in practice: competition will not hurt you

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Received 17 October 2002; received in revised form 10 February 2004; accepted 9 March 2004

Abstract

Since the introduction of school vouchers in 1992, independent and public schools in Sweden operate on equal terms. We analyze the effects of competition on the public schools using data on the results of 28,000 ninth graders. Because the decision on which school to attend is a choice variable, sample selection models are used. To account for the potential endogeneity of the share of students attending independent schools, we use instrumental variable estimation. We also estimate panel data models on 288 Swedish municipalities. Our findings support the hypothesis that school results in public schools improve due to competition.

JEL classification: H42; H52; I28; L19
Keywords: Public education; Independent schools; Student achievement; School vouchers; School choice; Sample selection model; Panel data model; Instrumental variable estimation

1. Introduction

The role of independent schools has been an important focus in the debate on the quality of schooling. A central issue is whether the use of public funds to finance privately run schools, through voucher schemes or charter schools, may enhance education. Several states and counties in the US have implemented limited experiments with such systems. In Europe, several countries have a long tradition of independent schools, that while subject to a varying degree of government regulation, also receive public financing to varying

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0047-2727/$ - see front matter
doi:10.1016/j.jpubeco.2004.03.004

degrees. However, to our knowledge, no country has implemented a more complete reform of school financing than Sweden did in the 1990s. Two parts of the reform are of particular interest: a voucher system which replaced the earlier centralized system of financing; and a parental choice reform.

The Swedish experience is of interest for at least three reasons. First, the reforms have been radical. Under the Swedish system, municipal and independent schools receive public financing on close to equal terms. Provided that they fulfill certain basic requirements, all kinds of schools are eligible, including religious schools and schools run by for-profit corporations. In this respect, Sweden differs from for example Denmark where private schools have received public funding for a long time, but where only parent-controlled, non-profit schools receive such funding. Further, the Swedish system applies to all children. This sets the reform apart from the so-called Milwaukee experiment (see, e.g., Rouse, 1998; Greene et al., 1999), which only provides vouchers to low-income groups. There are really only two serious limitations to the operation of independent schools. In order to receive public funds, they must pledge not to charge an additional tuition fee from the students. Obviously, this rules out competition on price. Further, the freedom in setting the rules of admission is limited: in particular, independent schools cannot refuse to accept low ability students.

Second, the country has experienced a rapid growth in the number of independent schools due to these reforms. The impact of the reform also differs between different municipalities[1] in Sweden. Enrollment in independent schools, at the compulsory school level, ranges between 0% and almost 20%. As Newmark (1995) points out in his criticism of Couch et al. (1993), studies of the effects of competition on public schools are in many cases unlikely to find any significant effects simply due to low variability in the data. He also questions the likelihood of competition from independent schools having any marked effect, unless the enrollment in independent schools varies over time. The high variability of private school enrollment in Sweden over time and space implies that these problems are somewhat mitigated.

Finally, unlike many other countries, most of the independent schools do not aim at any special group of students, such as those from a certain religious group. Rather, most of them are non-denominational and compete with public schools for the same group of students; something which is likely to make any effect of competition more noticeable. Further, the socioeconomic composition of students attending independent schools is not radically different from those attending public schools, thereby making inference easier.

Competition between schools raises several questions. Among the important issues are how public accountability may be ensured when public money is allocated to private institutions, and issues of equity and segregation. Of obvious interest is if the achievements of students in independent schools differ from those of students in publicly run schools. This has been an important research issue since at least the beginning of the 1980s, when Coleman et al. (1982) found evidence that private schooling was more effective. More recently, the so-called Milwaukee experiment, in which school vouchers are provided to some low-income students, has received considerable attention. Rouse

[1] In Sweden, schooling is primarily the responsibility of the municipalities, the lowest tier of government.

(1998) and Greene et al. (1999) find evidence that students who, under the Milwaukee voucher program, attend private schools score higher on some achievement tests than students in the public schools. Based on data from a randomized field trial in three US cities, Howell et al. (2002) find improved test scores among African American students (but not students from other ethnic groups) who attended private schools, relative to their public-school peers.

In this paper, however, we will focus on another issue, namely what effect competition from independent schools has on the public schools. There are two main reasons to select this topic. In the public debate that followed, the Swedish reform, the focus of criticism was that the public school system was depleted of resources, and thus impoverished as a result of the expansion of independent schools. While a lot of anecdotal evidence was presented in support of the beneficial effects of competition, several stories of public schools with depleted resources and diminished quality circulated. A former Minister of Education, Carl Tham, even claimed that "(t)here is inevitably a conflict between freedom of choice and a good school for everybody" (Tham, 2001). It could be argued that students that have selected to use the voucher system to attend an independent school, have demonstrated that they find this school superior. The main concern of policy makers should then be students that are still in the public school system. Even if freedom of choice improves the average results of students, it may still be undesirable if a considerable number are "left behind", as some critics of the system claim. In particular, this may be the case if socially or otherwise disadvantaged students choose independent schools to a lesser extent than other students.

An empirical analysis of competition between schools is necessary, since the theoretical predictions are unclear. On the one hand, public schools may lose the best teachers and the best students to independent schools which, in turn, may have an adverse effect on students remaining in public schools. Epple and Romano (1998) have argued that in the presence of a peer-group effect, i.e., a positive "learning externality" from sharing the classroom with more able students, low-ability students may be adversely affected by school choice. Hoxby (2001) and Hanushek et al. (2001) find empirical evidence supporting the existence of such peer group effects. On the other hand, increased competition and the risk of losing students and resources, give public schools incentives to improve education and may lead to more experimentation with regard to, e.g., pedagogical methods. In addition, competition may have a beneficial impact on teachers for two reasons. First, it may induce them to increase their work effort. Second, if public schools in effect form a monopsony for teachers' services, competition may result in higher salaries, which would attract able people to the profession. Both these effects have been documented empirically. Rapp (2000) finds that teachers work more diligently when school choice is introduced. Vedder and Hall (2000) and Hoxby (1994) find that teachers' salaries tend to rise due to increased competition from private schools.

Competition between schools has received increased attention among economists. Noting that the predominant form of competition between schools in the US is through Tiebout choice, Hoxby (2000) examines the effects of varying sizes of school districts on student performance. The underlying logic is that Tiebout choice is more effective when the number of jurisdictions in a given area is large. Her results suggest that competition through Tiebout choice enhances productivity in public schools. In fact, productivity increases due to changes in both the numerator and the denominator: student achievement improves, while the costs of schooling falls. Hoxby also finds that private school enrollment is lower where Tiebout choice is more intense. Plausibly, this is because the demand for private education is smaller where public schools are of high quality. Since school district boundaries may be endogenous—there is an incentive to merge poorly performing school districts with neighboring districts—Hoxby uses an instrumental variable estimation. An important methodological finding is that an estimation that does not take this endogeneity into account is biased towards finding no effect of Tiebout choice.

A few papers have explicitly examined competition between public and private schools, and are thus similar in spirit to the present paper. The main question asked in these studies tends to be if the achievement of students in public schools is improved when the proportion of students in private schools is higher. A central empirical issue in such studies is that the key variable, i.e., the number of students attending private schools, may be endogenously determined. If public schools are of poor quality, the demand for private schools may be larger. If this is not accounted for, the effects of competition may be underestimated. To address the endogeneity problem, Dee (1998) instruments for the private school share, using the population concentration of Catholics. The rationale for this is that since many private schools in the US are run by Catholic institutions and thus this variable will be highly correlated with the presence of private schools. Dee uses data on school districts in US 18 states, and finds that competition significantly improves high school graduation rates. Hoxby (1994) uses a somewhat similar strategy, but uses a richer set of denominational variables as instruments, and uses data on individual student achievements. Her conclusion is that competition from private schools has a significant positive effect on the quality of public schools in terms of educational attainment, measured as the highest grade completed by the age of 24, wages (also measured by the age of 24) and high school graduation rates. Couch et al. (1993) also find competition to positively effect public school performance, using the average scores in a compulsory mathematics test in all counties in North Carolina. However, this study has been criticized by Newmark (1995) who claims that the results are not robust.

The central contribution of the present paper is that we use an extensive data set to study the effects of a truly radical reform of school financing. Before the reforms began in the early 1990s, Sweden had a completely centralized school system with school financing mainly provided by the national government. Hardly any independent or private schools existed. Today, Sweden has more generous rules for the use of public funds to finance independent schools than any other country, with the possible exception of the Netherlands. We use a data set of around 28,000 ninth graders in public and independent schools in the scholastic year 1997/1998, containing information on grades, test results and socioeconomic background variables. Our approach differs somewhat from previous research in that we use sample selection models in order to simultaneously model both the students' choice of school and their educational results. This approach is used to take account of the fact that students choosing public schools are not a random sample of all students. This is one of the two key identification problems we meet when we want to study how competition from independent schools affects public schools.

The second identification problem is caused by the fact that independent schools are not established by chance. In particular, the availability of independent schools may be a function of the quality of public schools, causing the key explanatory variable to be endogenous. A additional complication is that the use of variables on religious affiliation as instruments for the share of independent schools is clearly not appropriate in Sweden. First, only a minority of the independent schools are denominational. Second, around 85% of the population belong to the Lutheran Church of Sweden,[2] and there is little regional variation. Third, religious minorities in Sweden differ from the majority population with regard to important socio-economic and cultural characteristics, which makes religious variables unsuitable as instruments. Instead, we use a number of political variables to construct an instrument. The motivation for this approach is that if the local authorities are hostile to independent schools, which in some cases they are, this may limit the expansion of such schools.

A further empirical complication is that unobserved heterogeneity between municipalities or between schools may exist. Since the data on individuals are from one year only, any panel data estimation is ruled out. However, robust standard errors of the coefficients are estimated. To complement the analysis of the data on individuals, we also run a separate set of regressions on data from all of Sweden's municipalities for the years 1993–1997. We estimate panel data models with the average grades in the municipalities as the dependent variable.

We find that the extent of competition from independent schools, measured as the proportion of students in the municipality that go to independent schools, improves both the scores on a national standardized mathematics test and the grades in public schools. This is confirmed by the results from the panel data models. The improvement is significant both in statistical and quantitative terms. This result holds for test results, final grades and for the likelihood that a student will leave school with no failing grades. Thus, our results confirm findings from earlier research which indicate that competition is beneficial for students in public schools. There is no indication that the expansion of independent schools has increased total expenditures on schools. Thus, the improved results imply that productivity has increased in the public schools.

The remainder of the paper is organized as follows. Section 2 treats the Swedish reforms, and what has happened since these were enacted. In Section 3, the data and the empirical analysis are presented. Section 4 concludes.

2. The Swedish experience

In Sweden, compulsory schools have traditionally been the responsibility of municipalities (the lowest tier of government). However, the municipal schools operated under strict national rules and regulations, and received funding from the national government. They also had to follow a national curriculum. Only a few independent schools, approved

[2] The Church of Sweden was not disestablished until the year 2000, which accounts for the high membership rate.

by the government, received government funding. In 1990, the system was altered and the municipalities were given greater authority over their own schools, and were also given full financial responsibility for the school system. In 1992, a new reform was implemented with the intention of giving independent schools funding on terms equal to those of municipal schools. Under the new law, municipalities were obliged to give funding to independent schools, on a per capita basis, amounting to 85% of the municipality's costs for schooling per student. The 85% rule was seen to be necessary not to put the municipal schools at a disadvantage, since the municipalities would still have to account for various administrative and overhead costs related to the municipalities' overall responsibility for the school system.[3] Parents were also given the right to choose school for their children. Similar reforms have been instituted for the upper secondary schools ("gymnasium"—roughly equivalent to high school). However, this paper mainly deals with compulsory schooling, which in Sweden is 9 years, from age 7 to 15.

The independent schools must be approved by the National Agency for Education to receive funding. There are some provisions for approval. The schools must meet certain quality requirements, and must work in line with the targets set for the compulsory educational system. They must also be open to all children. Thus, they may not base admission on ability or on religious or ethnic origin. Finally, they are not allowed to charge tuition. Among the approved schools are schools owned by teacher or parent cooperatives, non-profit organizations and privately owned firms. The municipalities are allowed to give an opinion on whether they consider the establishment of an independent school to be harmful to existing schools, and their views are taken into account by the National Agency for Education. However, the municipalities have no veto, and are bound by law to finance an independent school once it has been approved. On several occasions, the Agency has approved schools against the will of the municipalities. While municipalities have no formal authority to stop the establishment of independent schools, the attitude of the municipal authorities is important in practice. As we will see below, such attitudes may affect the likelihood that an independent school is established.

The 1992 reform has had a drastic effect on the number of independent schools. In 1991/1992, Sweden had 90 independent schools at the compulsory level, a number which had increased to around 400 in the academic year 2001/2002. Still, the number of independent schools is small compared to the total number of schools, which is about 5000. The number of students in independent schools is a very small fraction of the total number of students, around 4%. However, this share is rapidly increasing, as enrollment in independent schools has grown by 10—12% a year in the past few years. Further, there is considerable variation between different municipalities. While around half of Sweden's municipalities have no independent schools, in several municipalities, around 10% or

[3] Such costs include, e.g., some supervision over the independent schools. Also, while an independent school may determine how many students to accept, the municipality is obliged to provide schooling for all children. The grounds for calculating the grants given to independent schools were altered in 1995, including some costs not previously included, and instead reducing the percentage rule from 85 to 75. The rules were altered again in 1996. It is a matter of debate whether the independent schools are over- or undercompensated, in comparison with municipal schools, and if the changes in the grant system made the system more or less generous. A government committee that undertook to explore this issue reached no firm conclusions (SOU, 1999:98).

more of the students attend independent schools: the largest share being above 18% for the academic year 1999/2000.

While religious or ethnic origin may not be a requirement for admission, there is no rule against denominational schools, or schools with a focus on specific ethnic groups. Muslim and Jewish schools have been approved, as well as Christian schools of various denominations. About 15% of the independent schools are denominational. However, the majority are either "general" schools (30%) or schools applying some distinct pedagogical idea, such as Montessori, Waldorf, Freinet or Reggio Emilia (30%). The remaining 25% are ethnic schools, schools with teaching in another language than Swedish or schools with a focus on some special subject, e.g., artistic schools, etc. The largest number of independent schools are found in the main urban areas, but several schools have also been started in rural areas and in fact, some of the municipalities with the largest share of students in independent schools are to be found in the sparsely populated northern parts of Sweden.

3. Empirical analysis

3.1. Model specification

Since the objective of this paper is to analyze how competition from independent schools has affected the quality of public schools, the dependent variable should be some measure of student performance. We will discuss how this performance variable is defined below. The key explanatory variable should be a gauge of the degree of competition from independent schools. As such a gauge, we use the share of students attending independent schools on each "market". Each municipality is defined as a "market", since students almost invariably attend schools in their own municipality. If the municipal schools are considered to be a single "firm", this variable is in fact one minus the one-firm concentration ratio. It should be noted that this variable is not calculated from our data. Instead, it is a figure for the independent school share in the municipalities in the entire compulsory school system, i.e., grades 1 through 9. We will discuss below if it is appropriate to treat all municipal schools as one firm.

We can formulate the "main" equation in our analysis as:

$$y_i = \beta_1 S_i + \boldsymbol{X}_i \boldsymbol{\beta}_2 + \varepsilon_i \tag{1}$$

where y_i is a measure of student i's performance in public schools, S_i is the share of students attending independent schools in the municipality where student i lives, $\boldsymbol{X}_i$ is a vector of other explanatory variables, β_1 and $\boldsymbol{\beta}_2$ are the parameters to be estimated and ε_i is an error term. The elements of vector $\boldsymbol{X}_i$ are characteristics of the individual, the municipality and the school, which will be discussed further below.

A central empirical challenge is to take account of the potential endogeneity of S_i. The demand for alternatives to public education is likely to be greater if public schools are of poor quality. In the Swedish setting, there may be a counteracting effect on the supply side. The entitlement to an independent school is determined on the basis of the cost of schooling in the municipality where it operates. Depending on the policy of the municipality, this entitlement may be calculated as the average over all students, or alternatively consist of a basic entitlement and additional funding for students with special needs. In the former case, funding may not cover the full costs if the school attracts a higher than average proportion of special-needs students. Since independent schools are not allowed to "top-up" by charging tuition fees, the financial viability of an independent school will depend on the policy of the municipality, and on how "costly" the students are. Since independent schools may not refuse to accept "high-cost" students, there are factors on the cost side that the individual school have difficulties in controlling. As a consequence, independent schools may not be viable in some municipalities. The likelihood that this should happen may be higher if the proportion of special-needs students is higher, and may thus be correlated with student performance.

In conclusion, our key explanatory variable, the share of students attending independent schools, may be a function of our dependent variable, students' educational achievement. To solve the resulting identification problem, we must find some instrumental variables that are correlated with the share of students attending independent schools, while they do not have an independent influence on students' grades.

In essence, our approach for handling the endogeneity issue resembles that of Hoxby (1994) and Dee (1998). We begin by estimating an equation explaining the share of students attending independent schools. These estimates are then used to construct an instrument for this variable that can be used when estimating the effects of competition on student performance in public schools.

We can formulate an equation explaining the share of independent schools in a municipality in the following way:

$$S_j = \boldsymbol{R}_j \boldsymbol{\alpha} + \xi_j \tag{2}$$

where S_j is the share of students attending independent schools in municipality j, $\boldsymbol{R}_j$ is a vector of explanatory variables, $\boldsymbol{\alpha}$ a vector of parameters to be estimated and ξ_j is an error term. We use j as an index instead of i to emphasize that, unlike in Eq. (1), the unit of observation is a municipality and not an individual. Since this equation is a reduced form expression, $\boldsymbol{R}_j$ should include variables expected to influence both the demand and the supply of education in independent schools. Endogeneity is caused by the fact that we would like to include the municipality equivalence of y_i, the dependent variable in Eq. (1), e.g., average grades in the municipality, as an element in the vector $\boldsymbol{R}_j$.

While Hoxby (1994) and Dee (1998) use religious variables as instruments for the share of independent schools, this is not likely to work in the Swedish setting, due to the reasons explained above. The rationale for our approach is that the attitude of the municipalities towards independent schools will influence the likelihood of such schools being established. While the municipalities cannot veto the establishment of independent schools, the National Agency for Education asks for their opinion. There are also informal ways in which a municipality may aid or hinder the establishment of an independent school, e.g., by delaying the necessary permits for the use of buildings by a school. Some municipalities also go out of their way to inform parents about independent as well as municipal schools, while others in effect try to discourage parents from choosing independent schools.

Naturally, it is hard to directly measure this "attitude". However, we may use the municipalities' policies in areas other than schooling as a proxy. In particular, it seems likely that the extent to which municipalities contract out their responsibilities is an indicator of their attitude to the "privatization" of public sector activities. While some Swedish municipalities perform virtually all their tasks in-house, many have contracted out their responsibilities to a considerable extent. At the same time, the extent to which non-school activities are contracted out should not have any independent effect on the educational achievements of school children in the municipality. We use data on the share of municipal activities contracted out in five areas: "infrastructure", i.e., road maintenance etc.; child care; care for the elderly and disabled; social services, i.e., treatment of drug addicts, aid to dysfunctional families, etc.; and finally, "business activities".

Since the reforms of school financing came into force in 1992, it seems safe to assume that the grades given in that year have not been affected by any change in the degree of competition from independent schools. Thus, it should be safe to include the average grades in the municipalities for the year 1992, which is the first year for which data are available, as an explanatory variable in Eq. (2).

We based the decision on which other variables to include in Eq. (2) partly on information from the National Agency for Education (2000). The agency has studied how the composition of students in independent schools compares with the student population at large. Their findings indicate that students in independent schools more often have an immigrant background. Also, the parents of students in independent schools on average have higher incomes and higher educations. We thus include the share of immigrants, the share of the population with no higher education, and a measure of average income in the municipality as explanatory variables. A partial explanation for the higher proportion of immigrants in independent schools may be that some independent schools have a special focus on minority groups. However, also when we exclude this category of independent schools, the pattern holds.

We also include a dummy variable indicating whether the municipality is situated in a major urban area, and a measure of population density. As such a measure, we use "population distance", which is calculated by Statistics Sweden, and which is a measure of the hypothetical average distance between inhabitants in the municipality under the assumption that they are evenly distributed. Thus, the value of this variable is higher, the more sparsely populated is the municipality. It is not obvious what signs the coefficients of these variables should have. It may be easier to start an independent school in a densely populated area, but on the other hand, the demand for an independent school may be greater if the closest public school is far away. In fact, some independent schools have been started as a direct result of the closing of rural schools.

In addition to the policy variables discussed above, we include a variable supposed to measure the resources the municipality devotes to schooling—the average cost per student. Strictly, this variable may also be endogenous, but we will ignore this complication. It is unclear what sign it should have in a reduced form equation such as Eq. (2). The demand for independent schools could be lower if municipal spending on schools is high. On the other hand, it may be easier to start an independent school when the financial conditions are advantageous.

Finally, two political variables are included in Eq. (2): the share of votes received by the non-socialist parties in the general election of 1998; and a variable indicating if the municipal government is non-socialist. The rationale for including these two variables is that the political views of the inhabitants of a municipality may affect the demand for independent schools. Since the main opposition to independent schools originates from the left of the political spectrum, we would expect the demand for alternatives to the public school system to be larger the larger the non-socialist share of the electorate.

The share of students attending independent schools is zero in several municipalities, but cannot be less than zero. To take this into account, Eq. (2) is estimated as a Tobit model.

In essence, the potential endogeneity of the share of students in independent schools is caused by the fact that independent schools are not established by chance. However, given the size and number of independent schools, self-selection remains a problem, since students are not randomly distributed between independent and public schools. Rather, this distribution is determined by the preference for independent schools. If this preference is related to the students' aptitude then not accounting for self-selection may lead to erroneous conclusions. If, for instance, more able students have a higher propensity to select independent schools, we may fail to find an effect of competition on student results in public schools, or even find an adverse effect.

We address this problem by using Heckman's approach. In other words, we treat y_i of Eq. (1) as a latent variable that is only observed if the student chooses to go to a public school. We may formulate a selection equation as:

$$w_i = \boldsymbol{Z}_i\gamma + \zeta_i \tag{3}$$

where $\boldsymbol{Z}_i$ is a vector of explanatory variables, γ is a parameter vector and ζ_i is an error term. The latent variable y_i of Eq. (1) is thus only observed if $w_i > 0$. The dependent variable in this model, w_i, may be viewed as the individual's preference for municipal schooling. Note that w_i is also a latent variable. What we observe is instead the binary variable:

$$w_i^* = \begin{cases} 1 \text{ if } w_i > 0 \\ 0 \text{ if } w_i \leq 0 \end{cases} \tag{4}$$

where $w_i^* = 1$ implies that the student attends a municipal school.

Our model can be summarized by Eqs. (1), (2) and (3). While it may be of interest to analyze why independent schools are established in some municipalities but not in others, our main reason for estimating Eq. (2) is to construct an instrument for the key variable in Eq. (1), which is our "main equation".

In the main equation, we use four different achievement variables as dependent variables. Measuring educational achievement is difficult. Obviously, we would like a measure that is as encompassing as possible. With this criterion, the "credit value" seems ideal. This value is calculated from the student's final grades in his 16 "best" subjects, and constitutes the basis for acceptance to high school. Thus, this value is roughly equivalent to a grade point average. On the other hand, there is a risk that this measure is not objective since it involves the subjective judgements of teachers. In particular, a possible effect of competition between schools may be to induce teachers to be more generous when grading students. The most objective measure we were able to find was the results on two of the five sub-test in the standardized national achievement test in mathematics, sub-tests A and B: Sub-test A tests the students' ability to comprehend mathematical symbols and expressions while sub-test B consists of short algebraic problems. This should limit the scope for subjectivity as far as possible. These tests are compulsory, and identical throughout the country.[4] We also used the results from the mathematics tests to test whether competition from independent schools inflated the grades of students. No such effect was found.[5]

The use of our last achievement variables is motivated by a different concern. It is conceivable that competition increases average performance but still hurts low-performing students. To explore this possibility, a dichotomous variable indicating if the student obtained passing grades in all of the three "cardinal" subjects (mathematics, Swedish and English), which the student must pass to be accepted to enter high school, was used. The purpose of including this variable is to have a measure of student achievement that will only improve if the results of low-ability students improve. In the regressions using this dependent variable, the econometric specification must be changed since we do not observe y_i of Eq. (1). The model is estimated as a bivariate probit model with partial observability of one of the variables.

As discussed above, our key explanatory variable in the main equation (Eq. (1)) is the share of students attending independent schools, which is supposed to be a measure of the degree of competition faced by municipal schools. However, since students are also allowed to choose between different municipal schools, a certain amount of competition will also exist between these. This is hard to measure but, ceteris paribus, the competition between schools, municipal or independent, will be fiercer the closer these are located, since students are less likely to make an active choice between schools if the distance between the closest and the second closest school is large. As a proxy for this factor, we use the population distance variable, and a dummy variable indicating if the municipality is located in a major urban area.

Other explanatory variables were included partly based on research by the National Agency for Education. The Agency's investigations indicate that the students' sex (female = 1), immigrant background, parents' educational level and income, and the

[4] Another advantage of using sub-tests A and B is that we have access to the exact scores of each student, thus giving us an approximately continuous dependent variable. (The maximum scores are 30 and 45, respectively.) For the three other mathematics sub-tests, only the grades, on a four-step scale, are reported. As an informal robustness test, we ran ordered probit regressions on the three other sub-tests, ignoring self selection and endogeneity. Using the robust standard errors, the competition variable was significantly different from zero at the 5% level in all cases but one, the oral part of the mathematics test. We also ran ordered probit regressions on grades in mathematics, Swedish and English, again ignoring self selection and endogeneity. This time, the competition variable is significant for the mathematics grade, but not for the English and Swedish grades. In all cases, the coefficient estimates are positive, however. The results are not reported but can be obtained from the corresponding author.

[5] The regressions testing for grade inflation due to competition are not reported, but may be obtained from the corresponding author.

number of students at the school influence student results. (Small schools have slightly better results) (Skolverket, 1998). We do not have access to family income, and thus have to substitute a municipal average. We also include a school spending variable which is defined as municipal spending per student excluding rental costs, since such costs are difficult to define due to varying accounting practices among municipalities. Also, the cost of renting a school building is unlikely to have any effect on educational results since rental costs per square meter vary considerably between different parts of the country. Our choice of explanatory variables is consistent with previous research on the effects of competition on educational achievement (e.g., Hoxby, 2000; Couch et al., 1993; Dee, 1998).

Some variables included in the selection equation (Eq. (3)) also figure in the main equation. Those are the variables for sex, immigrant background, parents' educational background, average income in the municipality, population distance, the dummy variable for major urban area and the municipality's educational spending per child. In addition, we include in the selection equation the vote share of the non-socialist parties in the general election of 1998 and a dummy variable indicating if the municipality has a non-socialist governing majority. It could be expected that the propensity to vote for a non-socialist party is correlated with the propensity to put your child in an independent school.

It is likely that the error terms are more closely correlated for individuals attending the same school than for individuals in different schools. To allow for this, we estimate robust standard errors, allowing for a cluster effect, using the procedure suggested by Rogers (1993).

3.2. Data

The empirical analysis uses four sets of data. One of these is used for the panel data analysis, and is further discussed in Section 3.4. The three other data sets are: data on around 28,000 youths; data on public and independent schools; data on municipalities.

The data on individuals have been assembled by the National Agency for Education and consist of socio-economic variables, grades and results on the national achievement tests for all students in the ninth grade in 34 Swedish municipalities for the scholastic year 1997/1998. The grades are reported both as a "credit value", which is a summary measure of all grades (roughly equivalent to a grade point average) and as grades in individual subjects. The national achievement tests, which are compulsory and standardized, are given in the subjects English, Swedish and mathematics. The students' grades on each sub-test, as well as on the test as a whole, are reported. For some of the sub-tests, the exact scores are reported, thus providing even more detailed information. The socio-economic variables include the parents' educational level, the student's sex and information on whether he/she or either of his/her parents is an immigrant. Parts of this data set have been provided by Statistics Sweden to the National Agency for Education. This is the case for all data on grades, such as credit value, and most of the socio-economic background data. Data on test results, however, have been collected by the Agency from schools in the municipalities included in the sample. Unfortunately, there is a considerable number of missing observations in this part of the data set, as will be discussed further below. Since the variable for the student's sex has also been assembled from the test database, there is also a large number of missing observations for this variable.

The school database contains variables indicating the type of school, the number of students, etc. The data on municipalities, finally, provide information on population distance, a measure of average income, the municipality's costs for the compulsory school per student and an indicator of whether the municipality is situated in a major urban area. Naturally, a key variable is the share of students attending independent schools. We also have information on a number of political variables. We have data on voting behavior in the 1998 election, the political affiliation of the municipal government and policy variables dealing with the degree to which municipal responsibilities have been contracted out. Data on schools were provided by the National Agency for Education while the municipal data are from Statistics Sweden.

The full data set covers 29,335 students in 34 Swedish municipalities. After excluding students attending some "odd" schools, such as hospital schools and a few schools run by the national government or regional governments, the data set consists of 28,065 students, 26,656 of which attend municipal schools and 1409 attend independent schools. Descriptive statistics for these are presented in Table 1.

The school database lacks data on one municipality, leaving us with a sample of 33 municipalities and 27,996 students, 26,587 of which attend municipal schools. As can be seen in Table 1, only a few observations are lacking in the data dealing with the students' grades (credit value, no failing cardinal subject and no failing grade), while almost 13% of the observations lack information on test results from sub-test B, and almost as many for sub-test A. It is not likely that the missing observations are random. In fact, the average credit value for the observations missing information on test results is 146, while the mean over all observations is 199. The situation is complicated by the fact that we also miss observations on the students' sex for observations lacking information on test results. Since sex is included as an explanatory variable in all regressions, these observations will have to be omitted in all regressions. Thus, it is essential that we test if our results are robust to different assumptions about the missing observations.

Also, while the average test scores are higher for students in independents schools, the average credit value is lower. The standard deviation for the credit value is also considerably larger for students in independent schools than for students in municipal schools. The divergence seems to be caused by a larger share of students with credit value zero in the independent schools. If observations with credit value zero are excluded, the average credit value is 202 for the municipal schools and 228 in the independent schools. We will comment on the implications of this when we discuss our results.

Eq. (2) is estimated using data on all Swedish municipalities. Descriptive statistics for the variables included in that regression are presented in Table 2. In 1998, Sweden had 288 municipalities. Missing observations leaves us with a sample of 280. Of these, 177 had at least one independent school in 1998. The data on grades are from the year 1992, the year the reforms were instituted. Thus, it seems reasonable that these grades have not been affected by the presence of independent schools. In 1992, grades were given on a five grade scale. The numbers in the table represent averages for all students in each

Table 1
Descriptive statistics, variables in Eqs. (1) and (3)

Variable	Minimum	Maximum	By municipality			By individual (all)			By individual (municipal school only)			By individual (independent school only)		
			Valid observations	Mean	Standard deviation	Valid observations	Mean	Standard deviation	Valid observations	Mean	Standard deviation	Valid observations	Mean	Standard deviation
Full sample			34			28,065			26,656			1409		
Credit value	0	320				27,456	199.1	65.24	26,075	199.6	62.83	1381	188.7	99.91
Sub-test A	0	30				25,000	19.95	6.934	23,734	19.82	6.941	1266	22.48	6.300
Sub-test B	0	45				24,543	21.57	9.266	23,293	21.37	9.233	1250	25.31	9.089
No failing Coeur subject	0	1				26,867	0.9167	0.2763	25,726	0.9155	0.2782	1409	0.9560	0.2052
No failing grade	0	1				27,219	0.7799	0.4143	26,075	0.7804	0.414	1144	0.7684	0.4221
Independent school share	0	10.2	33	2.412	3.108	27,996	5.221	3.433	26,587	5.079	3.431	1409	7.889	2.110
Woman	0	1				26,079	0.4983	0.5000	24,695	0.4969	0.5000	1384	0.5231	0.4996
Immigrant background	0	1				27,111	0.2084	0.4062	25,923	0.2048	0.4035	1188	0.2879	0.4530
Parents' educational background	1	3				26,087	2.381	0.6696	24,758	2.367	0.6715	1329	2.637	0.5753
Income (municipal, 100 SEK)	816.1	1197	33	952.1	90.34	27,972	1013	100.9	26,587	1010	100.4	1385	1060	97.71
Urban dummy	0	1	33	0.2121	0.4151	27,996	0.5108	0.4999	26,587	0.4989	0.5000	1409	0.7339	0.4421
Population distance	17	1077	33	222.6	224.5	27,996	94.30	107.3	26,587	96.46	108.5	1409	53.52	70.28
No. of students in school	1	226	33	105.4	30.46	27,972	116.5	41.72	26,587	119.3	39.17	1385	62.61	51.26
School cost	34,600	54,900	33	42,310	4791	27,996	44,420	5109	26,587	44,300	5091	1409	46,807	4856
Non-socialist municipal government	0	1	33	0.2121	0.4151	27,996	0.0890	0.2848	26,587	0.0925	0.2898	1409	0.0227	0.1490
Non-socialist vote share	0.2359	0.5397	33	0.4040	0.0892	27,996	0.4611	0.0701	26,587	0.4592	0.0708	1409	0.4970	0.0424

For variables defined at the municipal level, descriptive statistics are presented both unweighted and weighted by the number of students in the sample from each municipality. For all variables, descriptive statistics are presented both for the entire sample, and for the sub-sample of students attending municipal schools.

Table 2
Descriptive statistics, variables in Eq. (2)

Variable	Minimum	Maximum	Valid observations	Mean	Standard deviation
Independent school share	0	14.6	288	1.610	3.108
Average grades 92	2.98	3.51	285	3.194	0.08717
Contracting (infrastructure)	0	50	288	7.830	9.369
Contracting (child care)	0	29	288	5.257	5.129
Contracting (elderly and disabled)	0	84	288	8.396	9.196
Contracting (social services)	0	55	288	17.17	10.45
Contracting (business)	0	81	287	12.26	15.04
School cost	29,200	58,900	284	41,385	4601
Share with immigrant background	0	9	288	2.191	1.390
Share without higher education	7	39	288	26.91	5.592
Income (municipal, 100s of SEK)	741	1755	288	956.2	121.4
Non-socialist municipal government (dummy)	0	1	288	0.3194	0.4671
Non-socialist vote share	0.1897	0.8412	288	0.4367	0.1095
Urban dummy	0	1	288	0.1319	0.3390
Population distance	17	2045	288	276.4	267.1

Note that these data cover all of Sweden's 288 municipalities.

municipality. In Table 3, the correlation between the share of students in independent schools and the five variables describing the degree to which municipal responsibilities have been contracted out is presented. This correlation is positive, and in some cases highly so, between the independent school share and the contracting-out variables. Thus, the first requirement on an instrumental variable, that it is correlated with the variable for which it is an instrument, seems to be fulfilled.

The other requirement on an instrument is that it should be exogenous. In the present case, our instruments would not be legitimate if they had a separate influence on students' achievement, apart from the influence through the effect on the number of independent schools. A test of this condition was performed by regressing student grades in 1992 on the other explanatory variables of Eq. (2). In this regression, none of the contracting-out

Table 3
Correlation between the contracting-out variable, and the independent school share

	Independent school share	Contracting (infrastructure)	Contracting (child care)	Contracting (elderly and disabled)	Contracting (social services)
Independent school share	–				
Contracting (infrastructure)	0.144	–			
Contracting (child care)	0.435	0.113	–		
Contracting (elderly and disabled)	0.385	0.138	0.384	–	
Contracting (social services)	0.129	0.074	0.083	0.208	–
Contracting (business)	0.151	0.386	0.221	0.303	0.088

The table displays the correlation between the degree to which the municipalities have contracted out their responsibilities and the independent school share.

Table 4
Explaining the share of independent schools

Independent school share	
Valid observations	280
Left-censored observations	103
Uncensored observations	177
Average grades 92	− 4.518* (2.443)
Contracting (infrastructure)	0.02579 (0.01990)
Contracting (child care)	0.1315*** (0.0428)
Contracting (elderly and disabled)	− 0.005913 (0.02513)
Contracting (social services)	0.01101 (0.01814)
Contracting (business)	− 0.009423 (0.01248)
School cost	3.73e − 05 (4.74e − 05)
Share with immigrant background	0.3868*** (0.1443)
Share without higher education	− 0.1338*** (0.04585)
Income (municipal, 100s of SEK)	0.0003759 (0.00249)
Non-socialist municipal government (dummy)	− 0.6807 (0.5038)
Non-socialist vote share	6.688** (2.69)
Urban dummy	1.764** (0.6853)
Population distance	− 0.0007302 (0.001036)
Constant	12.37 (8.187)
Standard error:	2.610 (0.1451)
Pseudo R^2:	0.1184

The table presents results from a Tobit estimation of Eq. (2), explaining the share of students attending independent schools in Sweden's 288 municipalities.

variables were significantly different from zero.[6] This seems to support the hypothesis that these variables are exogenous. However, it is conceivable that municipal policies that favor the establishment of independent schools are correlated with other policies that affect educational standards, and that are also correlated with the municipalities attitude towards the ‘‘privatization’’ of public sector activities. We are not able to test this hypothesis, but it is a caveat to keep in mind when interpreting our results.

3.3. Results

3.3.1. Endogeneity, self-selection and the preferred specification

The result from a Tobit estimation of Eq. (2) is presented in Table 4. Only one of the coefficients on the contracting-out variables, child care, is significantly different from zero at any usual level of significance. The variables on the immigrant share and the share of the population without higher education are also significantly different from zero. In the first case, the sign is positive, and in the second case, it is negative, thus indicating that the share of immigrants and people with higher education have a positive effect on the number of students attending independent schools. The coefficient on the vote share of the non-socialist parties is also positive and significantly different from zero, which confirms our suspicion that voting behavior will influence the likelihood that a student attends an independent school. However, the political affiliation of the municipal government has no

[6] The regression results are not presented, but are available from the corresponding author.

significant effect. The urban dummy is significant with a positive sign, which is not surprising, since a high proportion of the independent schools are established in the main urban areas in Sweden.

The most interesting result from this regression is perhaps that the coefficient on average grades in 1992 is negative, and significant at the 10% level. The share of students attending independent schools thus seems to be larger where school quality is low. This is in line with Hoxby's (1994) results, and indicates that we run the risk of underestimating any positive effects of competition if we do not take endogeneity into account. Our results give no support to the hypothesis that independent schools are more likely to be established in municipalities with "easy customers", i.e., few low-ability students.

For each of our four result measures, we use four different econometric specifications for the "main" equation, i.e., Eq. (1). Estimations (I) and (II) in each of the tables ignore self-selection, while models (III) and (IV) are estimated using Heckman's approach with Eq. (3) forming the selection part of the model. In the two cases where the dependent variable is dichotomous, the resulting equation is formulated as a probit. Models (II) and (IV) are estimated using an instrument for the share of students attending independent schools. This instrument is constructed using the estimates of Eq. (2) presented in Table 4. In all cases, the presented standard errors are from robust estimators allowing for clustering.

We should be cautious in interpreting the results for one of our school result variables, scores on sub-test A. (The estimation results are not presented, but are available from the corresponding author.) In this model, the estimate for ρ, i.e., the coefficient of correlation between the error terms in the selection equation is -1, which is at the lower bound for that coefficient. Most likely this indicates that the model is misspecified. In fact, while the distribution of test scores for sub-test B is nicely bell-shaped, the distribution of scores for sub-test A is markedly skewed to the right. As a measure of achievement, our Eq. (1), the score on sub-test A is in effect censored to the right, thus violating the assumptions behind the Heckman model. This is the result of the design of the mathematics test, where sub-test A is mainly intended to determine which students should get a passing grade, while the other sub-tests are used to determine which of the passing grades a student should get. We will ignore the regressions using sub-test A as the dependent variable while noting that the results support our conclusions.

Estimation results for the three remaining achievement variables are presented in Tables 6–8. We will first discuss which specification should be our preferred model. For each set of four specifications, we would first like to test the hypothesis that the "critical variable", i.e., the share of students attending independent schools is in fact exogenous against the alternative hypothesis that this variable is endogenous, and second, the hypothesis that Eqs. (1) and (3) are independent, against the hypothesis that they are not. For the first of these, we used a Hausman test. We could not in any case reject the hypothesis that the share of students attending independent schools is exogenous at any reasonable level of significance. Based on these tests, we should thus prefer the non-IV specifications. The hypothesis that Eqs. (1) and (3) are independent was then only tested for the non-IV specifications, and could be rejected at least at the 5% level for the three continuous variables, but not at any usual level of significance for the two dichotomous variables. In our analysis of the results, we will thus focus on the Heckman models (III) for the continuous variables, and on the univariate probit models (I) for the dichotomous variables, however, the results are similar in all econometric specifications. Test statistics are presented in Table 5a and b.

Let us first comment on the correlation parameter, ρ, in the Heckman models. Negative values of this parameter imply that students with low achievement, given observable variables, tend to self-select into the public schools, while positive values suggest the reverse. Disregarding the regression with sub-test A as the dependent variable, our estimates of ρ fall in a wide range: between -0.95 and 0.42. While the latter value, which is obtained in the regression with the dummy variable for students with no failing grades in the cardinal subjects as the dependent variable, is not significantly different from zero, it diverges considerably from the results in the other specifications. There is a plausible explanation for this. Recall from the above discussion about the descriptive statistics in Table 1 that a considerable fraction of the students in independent schools had a credit value equal to zero. However, if all individuals with a credit value equal to zero were excluded, the average credit value for students in independent schools were higher than for students in the public schools. This suggests that there is an overrepresentation both of high-achievers and of low-achievers in the independent schools. Thus, there may actually be negative self-selection when we consider the continuous variables and positive, or no self-selection, when we use the dichotomous variable since the latter will only be zero for students at the lower end of the achievement scale, and one over the major portion

Table 5

(a) Test statistics of Hausman tests of exogeneity of the independent school share variable

Variable	Hausman test statistic (significance level within parentheses)			
	Linear regressions (I vs. II)		Sample selection models (III vs. IV)	
Credit value	0.83	(0.9991)	2.53	(0.9801)
Sub-test B	6.16	(0.6293)	0.23	(1.0000)
No failing Coeur subject	0.19	(1.0000)	3.62	(0.8895)

For the three achievement measures, the hypothesis that the difference in the estimates in the non-IV and IV specifications was not systematic was tested. The test was performed for the linear specification and for the sample selection models. The test statistic, which under the null hypothesis is distributed χ^2, and the lowest level of significance at which the null hypothesis can be rejected, are presented in the table above. As can be seen, the null cannot be rejected at any ordinary level of significance.

(b) Test statistics of Wald test of independence of the selection and main equations

Variable	Test statistic (significance level within parentheses)	
Credit value	39.98	(0.0000)
Sub-test B	5.45	(0.0196)
No failing Coeur subject	0.76	(0.3835)

For the three achievement measures, the hypothesis that the selection equation and the main equation are independent is tested by testing the null hypothesis that the covariance of the errors in the two equations, ρ is equal to zero, using a Wald test. The test statistic, which under the null hypothesis is distributed Chi-2, and the lowest level of significance at which the null hypothesis can be rejected, are presented in the table above. As can be seen, the null can be rejected at least at the 5% level for the two continuous variables, but not at any usual level of significance for the dichotomous variable.

of this scale. This hypothesis is in line with the findings of the Government Committee on Funding to Independent Schools (SOU, 1999:98). It found that parents of children with special needs, such as learning disabilities, often choose an independent school when they did not consider that their children got the attention they needed in the public school.

Considering the results on the two other continuous variables, the estimated values of ρ in the credit value regression suggests much stronger self-selection (-0.95) than does the sub-test B regression (-0.23). This suggests that students who are relatively stronger in subjects other than mathematics choose independent schools to a higher degree. This could be a consequence of the fact that several independent schools specialize in subjects other than mathematics; most commonly foreign languages, music or the arts.

3.3.2. Main results: competition raises student results

Our main results are presented in Tables 6–8. In all of these regressions, the coefficient on the variable with the share of children attending independent schools is positive. In the preferred econometric specification (Model (III)) for the continuous dependent variables, it is significantly different from zero at the 1% level. The results thus suggest that competition improves the quality of public schooling. While the size of the coefficient varies between the different econometric specifications, the qualitative results remain the same. The coefficients are positive and significantly different from zero in all cases, including sub-test A.

In the specification with a dichotomous dependent variable, the results are less clear. While the coefficient is positive in all estimated models, it is significant in only one specification: the instrumental variable probit (II). Since this is not our preferred specification, based on the tests performed, we should probably not attribute too much importance to this result.[7] Our failure to get any clear results in these specifications could either be due to the fact that low-ability students benefit less from the increase in quality due to increased competition than the average student, or simply be a result of the lower amount of information contained in a dichotomous variable, as compared to a continuous result variable. However, the main point is that there is no evidence that low-achievers are adversely affected by increased competition from independent schools.

It is comforting to note that in all cases, the instrumental variable estimates are larger in magnitude than the comparable non-IV estimates. Thus, we are likely to err on the side of caution in claiming to find a positive effect of competition. Our rejection of the OLS model against the Heckman specification, and the fact that the estimated coefficients in the latter are larger in magnitude, implies that we are likely to underestimate the positive effects of competition if we ignore self-selection.

The estimation results on the other coefficients offer few surprises. Girls have significantly better school results than boys. Parents with a higher level of education have a positive influence on school achievements, while children from an immigrant background have significantly worse results. These coefficients are highly significant in all cases, usually at the 1% level. The coefficient on the income variable is positive, and significant in one case (sub-test B).

[7] In addition, this result is not robust to slight alterations of the model, such as minor changes in the list of explanatory variables. The coefficient is not significant, though still positive, if we exclude the population distance variable and the dummy for major urban areas.

Table 6
Explaining student results: credit value

Main equation	Dependent variable: credit value			
	I	II	III	IV
Uncensored observations	22,961	22,961	22,961	22,961
Censored observations	–	–	1098	1098
Independent school share	0.5381*	1.493**	1.461***	3.286***
	(0.2849)	(0.6773)	(0.5379)	(0.9745)
Woman	19.89***	19.90***	19.96***	19.99***
	(0.9012)	(0.9018)	(0.9522)	(0.952)
School cost	− 0.0003269	− 0.0003391	− 0.0003355	− 0.000317
	(0.0002117)	(0.0002065)	(0.0003082)	(0.0002972)
Immigrant background	− 7.098***	− 7.311***	− 4.134**	− 4.556**
	(1.49)	(1.491)	(1.919)	(1.932)
Parents educational background	29.18***	29.13***	31.15***	31.12***
	(0.752)	(0.7424)	(0.9427)	(0.9461)
Income (municipal, 100s of SEK)	0.009453	0.005358	0.01389	0.005293
	(0.009754)	(0.01012)	(0.01587)	(0.01612)
Urban dummy	1.839	0.4814	1.515	− 0.7564
	(2.410)	(2.709)	(3.735)	(4.073)
Population distance	0.01434*	0.02121**	0.01434*	0.02784***
	(0.007834)	(0.008180)	(0.008355)	(0.009086)
No. of students in school	0.04687**	0.04262**	0.04309**	0.03277*
	(0.01833)	(0.01854)	(0.01691)	(0.01718)
Constant	122.4***	124.0***	113.9***	116.1***
	(10.69)	(10.73)	(16.59)	(16.60)
Selection equation	Selected: students attending municipal schools			
	I	II	III	IV
Woman			− 0.1241**	− 0.1238**
			(0.05266)	(0.05421)
School cost			− 0.0000338	− 0.0000281
			(0.0000285)	(0.0000246)
Immigrant background			− 0.1549**	− 0.1555**
			(0.07849)	(0.0775)
Parents educational background			− 0.3714***	− 0.3692***
			(0.04171)	(0.04101)
Income (municipal, 100s of SEK)			0.003311	0.002515
			(0.002644)	(0.002206)
Non-socialist municipal government (dummy)			1.634**	1.41**
			(0.8273)	(0.6839)
Non-socialist vote share			− 9.899*	− 8.244*
			(5.537)	(4.503)
Urban dummy			− 0.1736	− 0.1408
			(0.2539)	(0.2272)
Population distance			− 0.001746**	− 0.001471**
			(0.0008738)	(0.0007029)
Constant			5.753***	5.467***
			(1.268)	(1.144)
ρ			− 0.9534	− 0.9525
			(0.02689)	(0.02841)

Table 6 (*continued*)

Selection equation	Selected: students attending municipal schools			
	I	II	III	IV
σ			54.77	54.76
			(0.7653)	(0.7624)
λ			− 52.22	− 52.16
			(1.571)	(1.653)

The table presents results from estimation of the main equation (Eq. (1)), with the credit value as the dependent variable. Estimations (I) and (II) ignore self-selection, while models (III) and (IV) are estimated using Heckman's approach, with Eq. (3) forming the selection part of the model. Models (II) and (IV) are estimated using an instrument for the share of students attending independent schools. This instrument is constructed using the estimates of Eq. (2) presented in Table 4. The presented standard errors are from robust estimators allowing for clustering. The preferred specification is model (III).

For the two demographic variables, the dummy for a major urban area and the population distance variable, only the latter is significantly different from zero, and only when the dependent variable is continuous. These variables were included to proxy for a possible effect of competition between different municipal schools. Our hypothesis was that competition between different municipal schools would be more intense, ceteris paribus, the closer they were located. Since the population distance variable is larger, the more sparsely populated the municipality is; this variable should have a negative sign to support our hypothesis. This is the opposite to our findings. However, while we find no evidence of a positive effect of competition between municipal schools, concluding that such an effect does not exist is probably premature. As pointed out above, our demographic variables are probably poor proxies for "intra-municipal" competition, and may also be correlated with unobserved factors that are important for school choice and educational achievement. An alternative specification, using a Herfindahl–Hirschman index, is discussed below in Section 3.3.3.

Turning to the remaining coefficients in the main specification of the model, we find that the school-cost variable is not significant, and that the sign is sometimes negative. While this result is consistent with the findings of Hanushek (1986) and others that the link is weak, or even nonexistent, between resources spent on education, and educational outcome, we would not like to draw any conclusions from this result. Since this is not the focus of the study, we have not taken the potential endogeneity of this variable into account.

The estimated coefficients on the number of students attending ninth grade in the school is positive, and significant in two cases. Thus, we find weak evidence that students in larger schools achieve better results, which is the opposite of what the National Agency for Education finds (Skolverket, 1998).

In the selection equation, we find that five coefficients are significant for both the credit value and sub-test B. The results are quantitatively the same as in an ordinary probit estimation of the likelihood that a student will attend a municipal school. Immigrant background and parents of a higher educational background reduce the probability that a student will attend a municipal school. Population distance is also negative and significant, indicating that people in sparsely populated areas are more likely to attend an independent school.

Table 7
Explaining student results: sub-test B

Main equation	Dependent variable: sub-test B			
	I	II	III	IV
Uncensored observations	21,815	21,815	21,815	21,815
Censored observations	–	–	1098	1098
Independent school share	0.1396**	0.2795**	0.1767***	0.2996*
	(0.05489)	(0.1335)	(0.05950)	(0.1582)
Woman	0.393***	0.3971***	0.4020***	0.4001***
	(0.1373)	(0.1375)	(0.1373)	(0.1371)
School cost	0.0000107	0.0000148	0.0000105	0.0000152
	(0.0000378)	(0.0000377)	(0.0000384)	(0.0000379)
Immigrant background	− 3.185***	− 3.214***	− 3.087***	− 3.187***
	(0.2151)	(0.2196)	(0.2262)	(0.2399)
Parents educational background	4.255***	4.252***	4.327***	4.273***
	(0.1074)	(0.1081)	(0.1168)	(0.1403)
Income (municipal, 100s of SEK)	0.004990***	0.00432**	0.005129***	0.004313 **
	(0.001755)	(0.001824)	(0.001793)	(0.001832)
Urban dummy	− 0.7592	− 0.9171*	− 0.7817	− 0.9331*
	(0.4717)	(0.5413)	(0.4825)	(0.5523)
Population distance	0.002197	0.003256**	0.002235*	0.003338**
	(0.001381)	(0.001532)	(0.001351)	(0.001535)
No. of students in school	0.003042	0.002333	0.003225	0.002328
	(0.00359)	(0.003736)	(0.003573)	(0.003744)
Constant	5.421***	5.482***	5.077**	5.382***
	(1.942)	(1.941)	(1.965)	(1.973)
Selection equation	Selected: students attending municipal schools			
	I	II	III	IV
Woman			− 0.03646	− 0.03386
			(0.04310)	(0.04343)
School cost			− 0.0000504	− 0.0000501
			(0.0000384)	(0.0000386)
Immigrant background			− 0.3142***	− 0.3074***
			(0.1010)	(0.1035)
Parents educational background			− 0.2731***	− 0.2761***
			(0.05880)	(0.05995)
Income (municipal, 100s of SEK)			0.005068	0.004989
			(0.003114)	(0.003098)
Non-socialist municipal government (dummy)			2.382***	2.355***
			(0.8483)	(0.8439)
Non-socialist vote share			− 14.2**	− 14.1**
			(5.948)	(5.936)
Urban dummy			− 0.2849	− 0.2857
			(0.3667)	(0.3684)
Population distance			− 0.00227**	− 0.002282**
			(0.001066)	(0.001078)
Constant			6.591***	6.617***
			(1.65)	(1.676)
ρ			− 0.2252	− 0.06626
			(0.09317)	(0.2497)

Table 7 (*continued*)

Selection equation	Selected: students attending municipal schools			
	I	II	III	IV
σ			8.520	8.493
			(0.06087)	(0.05504)
λ			− 1.918	− 0.5627
			(0.8009)	(2.122)

The table presents results from estimation of the main equation (Eq. (1)), with the scores on sub-test B as the dependent variable. Estimations (I) and (II) ignore self-selection, while models (III) and (IV) are estimated using Heckman's approach, with Eq. (3) forming the selection part of the model. Models (II) and (IV) are estimated using an instrument for the share of students attending independent schools. This instrument is constructed using the estimates of Eq. (2) presented in Table 4. The presented standard errors are from robust estimators allowing for clustering. The preferred specification is model (III).

The two "political variables" are both significant, with opposite signs. Taken at face value, this implies that where non-socialist parties have taken a higher share of the votes cast, there is an increased likelihood of students attending independent schools. However, given the vote share, this probability is reduced if the municipality has a non-socialist governing majority. This may seem surprising but a reasonable hypothesis is that the need to opt out of the public school system may be reduced if the governing majority is of the same political affiliation as the parent. The estimation results are consistent with the estimates of Eq. (2).

3.3.3. Additional tests[8]

To test the robustness of our results, several alternative specifications of the model were used. First, an alternative, and perhaps more natural way, to measure competition between municipal schools is to use a Herfindahl–Hirschman index. We re-estimated the model including this variable and found that for sub-test B, the coefficient is negative but not significant. In the credit value regression, the HHI gets a positive and significant coefficient, indicating that competition between municipal schools has a negative impact on results, but when the HHI is interacted with the urban dummy and the population distance variable the coefficient is no longer significant. Our conclusion is that the HHI is not a good way to measure competition between municipal schools, since the impact of such competition depends both upon the demographics of the municipality and on the policies followed by the municipal authorities.[9]

[8] Detailed results are not reported in this section, but are available from the corresponding author.

[9] To give a picture of the wide disparity of policies followed by the municipalities, the author checked with two municipalities what their policy was concerning the right to choose between different municipal schools. In Uppsala (HHI = 0.066, population 180,000), it took a number of phone calls to figure out the procedure, which was to contact the headmaster of the school to which the student wanted to transfer. A transfer was only granted if the school has surplus capacity, which it often does not have, since the capacity of the school is determined on the basis of its "catchment area". The decision is taken by the headmaster himself. The municipality of Nacka (HHI = 0.15, population 76,000), on the contrary, extended the voucher system to cover the municipal schools several years ago. Parents are encouraged to make an active choice and may do so over the Internet. It takes a few clicks on Nacka's homepage to find out how this is done. The information is available at http://www.nacka24.nacka.se/ in English. (As well as in Spanish, Finnish, Arabic and, naturally, Swedish).

Table 8
Explaining student results: no failing grade in the cardinal subjects

Main equation	Dependent variable: no failing grade in cardinal subjects			
	I	II	III	IV
Uncensored observations	22,930	22,930	22,930	22,930
Censored observations	–	–	1098	1098
Independent school share	0.01215	0.04745**	0.0009692	0.04133
	(0.008987)	(0.02228)	(0.01747)	(0.03595)
Woman	0.1869***	0.1871***	0.1801***	0.1857***
	(0.03039)	(0.03046)	(0.03194)	(0.03158)
School cost	3.94e − 06	2.42e − 06	4.16e − 06	2.36e − 06
	(7.42e − 06)	(7.26e − 06)	(7.47e − 06)	(7.23e − 06)
Immigrant background	− 0.3174***	− 0.3252***	− 0.3360***	− 0.3319***
	(0.04857)	(0.04910)	(0.05175)	(0.05213)
Parents educational background	0.5048***	0.5031***	0.4730***	0.4953***
	(0.02345)	(0.02343)	(0.05592)	(0.04341)
Income (municipal, 100s of SEK)	0.0002725	0.0001522	0.0002372	0.0001505
	(0.0003551)	(0.0003583)	(0.000368)	(0.0003567)
Urban dummy	− 0.05760	− 0.1140	− 0.04966	− 0.1078
	(0.08134)	(0.08833)	(0.08157)	(0.09135)
Population distance	0.0003766	0.0006337**	0.0003703	0.0006094**
	(0.0002554)	(0.000287)	(0.000254)	(0.0003082)
No. of students in school	0.002437***	0.002303***	0.002359***	0.002305***
	(0.0006161)	(0.0006209)	(0.0006131)	(0.0006195)
Constant	− 0.3800	− 0.3100	− 0.2739	− 0.2774
	(0.3768)	(0.3719)	(0.4288)	(0.4059)
Selection equation	Selected: students attending municipal schools			
	I	II	III	IV
Woman			− 0.03000	− 0.03286
			(0.04178)	(0.04214)
School cost			− 0.0000461	− 0.0000472
			(0.0000385)	(0.0000385)
Immigrant background			− 0.2931***	− 0.2986***
			(0.2931)	(0.1012)
Parents educational background			− 0.2840***	− 0.2855***
			(0.05942)	(0.05941)
Income (municipal, 100s of SEK)			0.004876	0.004909
			(0.003106)	(0.003106)
Non-socialist municipal government (dummy)			2.299***	2.324***
			(0.8694)	(0.8562)
Non-socialist vote share			− 14.13**	− 14.09**
			(5.941)	(5.949)
Urban dummy			− 0.3245	− 0.3071
			(0.376)	(0.3755)
Population distance			− 0.002393**	− 0.002346**
			(0.001069)	(0.001069)
Constant			6.641***	6.632***
			(1.638)	(1.652)
ρ			0.4157	0.1583
			(0.4017)	(0.5854)

The difficulties of drawing any conclusions from these regressions are aggravated by regional differences in grading. While we did not find any evidence of "grade inflation" due to competition from independent schools, we did find evidence of regional differences in grading practices. These results point to a potential source of bias in our estimates. Since the share of independent schools is different in different parts of the country this could affect our results. To test the robustness of our results, we tested a number of different specifications of the model, with regard to the urban dummy and the population distance variables. We tested to exclude these two variables, to include the logarithm or the square of the population density variable, and all combinations thereof. In all regressions, the qualitative results were the same as in the main regressions. For the credit value and sub-test B, the changes in the estimated coefficients were in the second significant digit.

School choice may have a beneficial effect on the average student but may also still hurt students with a poor socio-economic background. To test this possibility, we re-estimated the model on students in municipal schools whose parents had no education above the compulsory school level and on students with immigrant background. We also estimated the model on the subsample forming the intersection of these groups. Since the groups become small, we only estimated the OLS models, and only the models with a continuous dependent variable. In these regressions, the coefficient on the independent school share was considerably smaller, and in one case even negative, and nowhere significantly different from zero. Thus, there is some evidence that students from a lesser socio-economic background benefit less from competition than the average student; but perhaps the most interesting point is that there is no evidence that these groups of students are hurt by competition.

Aside from the points raised above, the robustness of our results may mainly be challenged for three reasons. First, there is a large amount of missing data, second, data are only from a selection (albeit a large one) of Swedish municipalities and, third, we could have included other explanatory variables.

To address the second of these issues, we tested to drop each municipality, one at a time, two at a time, and three at a time, for all possible combinations of the 33 municipalities, using credit value and sub-test B as dependent variables. We estimated the models using the preferred Heckman specification and OLS. The changes in the parameter estimates are in most cases in the second significant digit, even when three municipalities are dropped. When one municipality was dropped, the coefficient on the independent school share was positive and significant in all cases. Of the total 12,034 Heckman estimations, it remained positive and significant in 11,519 cases. In all but six

Note to Table 8:

The table presents results from the estimation of the main equation (Eq. (1)), with a dummy variable indicating if the student have passing grades in the three cardinal subjects (mathematics, Swedish and English) as the dependent variable. Estimations (I) and (II) ignore self-selection and are estimated as probit models, since the dependent variable is dichotomous, while models (III) and (IV) are estimated as bivariate probit models with partial observability and with Eq. (3) forming the selection part of the model. Models (II) and (IV) are estimated using an instrument for the share of students attending independent schools. This instrument is constructed using the estimates of Eq. (2) presented in Table 4. The presented standard errors are from robust estimators allowing for clustering. The preferred specification is model (I).

cases, it was significant for at least one of the two result variables. In most cases where the Heckman estimates did not result in a coefficient significantly different from zero, the coefficients were positive and significant when the models were estimated using OLS indicating that failure to get significant coefficient estimates is a result of estimation problems as the sample size decreases.

To test whether the large number of missing observations for test results, and for the dummy variable for the student's sex, has any serious consequences for our results, we replaced the missing values under a few different assumptions. First, the missing sex dummy was replaced by 0.5. We then replaced the missing values for the credit value and the student's grade in mathematics with zero, thus assuming that all students for which data were missing were extremely low ability students. Finally, we regressed the test results from sub-tests A and B, respectively, on the credit value and the mathematics grade, and also on each of these separately, and replaced the missing values for these tests with the predicted values. The values of the coefficients fell slightly but were still significant at the 5% and 1% level, respectively. As an additional test, we replaced all missing data for the test scores and for the credit value with zero, and ran the regressions on these three variables. This assumption is clearly extreme since many students with missing data on the test scores but with data available on grades have above average grades in mathematics. Under this assumption, the coefficients in the regressions with sub-test A and the credit value as dependent variables are smaller in value but still significant at the 1% and 5% levels of significance, respectively. For sub-test B, the value of the coefficient is higher, but the estimate is no longer significant.[10]

We have already discussed a number of alternative specifications of our regressions. However, two other points have been raised in comments on our results: that we should include different variables for the father's and the mother's level of education rather than having one variable for the household level of education and that we should include variables describing the population structure of the municipalities. We ran our model using three such variables, the share of the population on welfare, the share of immigrants, and the share of the population with no education above the compulsory school level. We also estimated models with all possible combinations of these three variables and the two definitions of the variables for the parents' educational level; in total fifteen combinations. The model was estimated with our three "main" achievement variables as the dependent variable. The alterations of the model had no noteworthy effect on the coefficient estimates.

Our conclusion is that the results are robust with regard to changes in the selection of the municipalities in the sample, and to the inclusion of different explanatory variables. It was only when extreme assumptions about the missing values were used that the qualitative results change. Thus, our results also seem to be robust in this respect.

[10] We also tested various other assumptions about the missing values for the variables on sex and parents' educational level: that all individuals with missing values are boys/girls and that their parents have only the lowest/intermediate/highest level of education. None of the results from these permutations of the model affected our conclusions.

Table 9
Results from the panel data models

	OLS	One-way		Two-way		Between
		FE	RE	FE	RE	
R^2	0.2481	0.7515		0.7632		0.3305
Adjusted R^2	0.2460	0.6880		0.7015		0.3210
Independent school share	0.0041 (0.0014)***	0.0037 (0.0017)**	0.0041 (0.0015)***	0.0019 (0.0020)	0.0037 (0.0016)**	0.0038 (0.0028)
Municipal school spending	2.1×10^{-6} (5.0×10^{-7})***	3.4×10^{-6} (5.7×10^{-7})***	2.8×10^{-6} (5.1×10^{-7})***	1.4×10^{-6} (7.0×10^{-7})**	1.6×10^{-6} (6.2×10^{-7})***	1.9×10^{-6} (1.1×10^{-6})**
Share without higher education	−0.0064 (3.7×10^{-4})***	−0.0026 (6.7×10^{-4})***	−0.0046 (4.8×10^{-4})***	7.2×10^{-4} 0.002	−0.0060 (7.6×10^{-4})***	−0.0070 (7.2×10^{-4})***
Share with immigrant background	−0.0058 (0.0015)***	−0.0085 (0.0029)***	−0.0076 (0.0021)***	−0.0013 (0.0033)	−0.0033 (0.0025)	−0.0051 (0.0030)**
Constant	3.3 (0.024)***	N/A	3.2 (0.028)***	3.1 (0.078)***	3.3 (0.035)***	3.3 (0.049)***

One, two and three asterisks indicate that the coefficient estimates are significantly different from zero at the 10%, 5% and 1% levels. The figures within parentheses are standard errors. The OLS and one-way specifications are resoundingly rejected against the two-way specification. (Test statistics are presented in Table 10a.) The one-way random effects model can be rejected against the fixed effects specification in a Hausman test at any usual level of significance. The two-way random effects specification, however, cannot be rejected at the 5% level against the fixed effects specification. (Test statistics are presented in Table 10b.) The models are estimated on data from all 288 Swedish municipalities for the years 1992 and 1994–1997.

3.4. Panel data analysis

In the three regressions described above, we allow for the error terms to be more closely correlated within observational units (schools) than across such units: In effect, we assume an error component model. An alternative specification would be to use a panel data setting. However, since our data on individuals are from one time period only the estimation of fixed effect models is not tractable. To check if non-observable heterogeneity is important in accounting for student results, we use a separate data set on municipal grade averages and background variables. Data are available for the years 1992 and 1994–1997 for all 288 of the municipalities that existed in 1997. Since two of these

Table 10a
Test statistics of likelihood ratio tests of two-way vs. one-way and OLS specifications

OLS vs. one-way	1584	(0.0000)
OLS vs. two-way	1654	(0.0000)
One-way vs. two-way	69.39	(0.0000)

For the panel data models, the one-way and two-way fixed effect specifications were tested against each other and against the classical linear regression specifications, using a likelihood ratio test. As can be seen from the table, the tests strongly favor the two-way specification.

Table 10b
Test statistics of Hausman tests of random vs. fixed effect panel specification

Hausman test statistic (significance level within parentheses)			
One-way (random vs. fixed)		Two-way (random vs. fixed)	
47.03	(0.0000)	9.24	(0.05530)

For the panel data models, the random effect specification was tested against a fixed effect specification, using a Hausman test. The test statistic, which under the null hypothesis is distributed χ^2, and the lowest level of significance at which the null hypothesis can be rejected, are presented in the table above. We may thus reject the one-way random effects specification against the one-way fixed effect specification, but cannot reject the two-way random effects model against the two-way fixed effect specifications.

were created in 1995, through some changes in administrative boundaries, the panel is unbalanced.

We estimate one-way and two-way models, using both fixed effects and a random effects specification. In Table 9, we also present estimation results from an OLS and a between model. Since this database contains aggregate data the variables are slightly different from the three previous models. It should also be noted that the dependent variable is an average for all students, i.e., for students in both municipal and independent schools. The variable indicating the share of students in independent schools is, however, the same as before. Apart from this variable, we include variables for the shares of the population with immigrant backgrounds and with no higher education. In addition, the municipality's spending on education (excluding rental costs) are included. This variable is also defined as before.

The preferred model is the two-way random effects model. In the two-way setting, we cannot reject the random effects model against the fixed effects model on the 5% level in a Hausman test, while we may reject the one-way specification against the two-way specification. (Test statistics in Table 10a and b.) However, the coefficients on the variable for the independent school share are close in all regressions, and significant in all specifications except the two-way fixed effect specification and the between model. The results from the panel data models thus confirm earlier results and seem to validate the error component setting assumed in the sample selection models.

Since the panel data models are specified as simple linear models, the coefficients are easily interpreted. To get an idea of the economic significance of increased competition we may compare the "resource coefficient", i.e., educational spending per student, with the impact of a larger percentage of students in independent schools. Such an analysis must obviously be taken with several grains of salt but gives some idea of the scale of the effect. Taken literally, the preferred model implies that an increase in the share of students attending independent schools by one percentage point would be equivalent to an increase in spending by about SEK 2000,[11] which corresponds to an increase in spending by over 5% (computed from the sample average).

[11] SEK 1 ≈ EUR 0.11 ≈ USD 0.13, as of December 2003.

4. Concluding remarks

The role of independent schools had been hotly debated long before Friedman and Friedman (1981) proposed school vouchers as a means of improving the quality of schooling. In the Netherlands, the "schoolstrijd" or "school war" in which advocates and opponents of state support to independent schools clashed was a central issue of political conflict during several decades around the turn of the century leading to a constitutional compromise in 1917 (Netherlands, 2004). In France, the role of Catholic schools was a highly divisive issue from the 1789 revolution and onwards (History of Education, 2004) and the establishment of secular schools was a central element of the anticlerical reforms of the Third Republic (Anticlericalism, 2004). In recent years, this debate has taken a new turn. Historically, much of the debate has focused on ideological issues, e.g., whether it is right to use public funds to support religious schools or if public schools should be used as a means of imposing greater national cohesion. In today's debate, however, efficiency motives have been introduced as an argument for school choice. It is claimed that increased competition between schools would be beneficial to educational quality.

A number of empirical studies have shown this argument to be valid. The present study further supports the finding that greater competition improves the standards of public schools. It is true, of course, that the fact that though our study indicates that student results are better with higher degrees of competition this does not necessarily imply that the better results are due to competition. However, the robustness of our results makes us feel justified in making this assertion. A widespread concern among opponents of school choice is that competition will hurt the public schools. The present study shows this fear to be without foundation.

Acknowledgements

Financial support from the Marianne and Marcus Wallenberg Foundation (Sandström) and the Expert Group of Public Finance, Swedish Ministry of Finance, (Bergström and Sandström) is gratefully acknowledged. We are grateful for valuable comments from Tore Ellingsen, Stefan Fölster, Sune Karlsson, Per Skedinger, Sören Wibe and seminar participants at the Stockholm School of Economics, the University of Uppsala, the IUI, the Institute of International Economics and the EEA Congress 2001, and from two anonymous referees.

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